

Executive Summary: This manual provides comprehensive instructions for installing the **Google Drive MCP** server inside ChatGPT, activating Developer Mode, configuring OAuth 2.0 endpoints, linking your Google Drive with secure single-use linking tokens, and utilizing all 32 Drive tools.

1. Prerequisites

- 1. ChatGPT Account**

ChatGPT Plus, Team, or Enterprise subscription with Custom GPTs / MCP app support enabled.
- 2. Google Account**

Any active Google Account (personal Gmail or Google Workspace) with access to Google Drive.
- 3. MCP Server Details**

Your server URL (e.g. `https://<domain>/mcp`) and OAuth Client ID & Secret from your admin.

2. Step-by-Step Installation in ChatGPT

STEP 1 — ENABLE DEVELOPER FEATURES IN CHATGPT

- 1. Open **ChatGPT** in your browser and log into your account.
- 2. Click your **Profile Icon / Name** in the bottom-left corner and select **Settings**.
- 3. Navigate to **Settings & Privacy** (or **Advanced / Developer Settings**).
- 4. Locate **Developer Mode** or **Plugin / MCP Developer Features** and toggle it to **ON**.

STEP 2 — ADD CUSTOM MCP CONNECTOR / ACTION

- 1. Navigate to **Explore GPTs** → Click **Create** (or go to **Settings → Connected Apps / MCP**).
- 2. Under the **Configure** tab, scroll down to **Actions / MCP Integrations** and click **Add MCP Server** (or **Create new action**).
- 3. Set the **Plugin / App Name**:
`Google Drive – [Your Name]` (e.g., `Google Drive – Alex` or `Google Drive`).
- 4. In the **MCP URL / Server Endpoint** field, enter your deployed MCP endpoint:
`https://<your-mcp-domain>/mcp`

STEP 3 — CONFIGURE OAUTH 2.0 CREDENTIALS

- Select **Authentication: OAuth** and enter the server OAuth configuration:
- **Client ID:** Enter your configured `CHATGPT_OAUTH_CLIENT_ID`
 - **Client Secret:** Enter your configured `CHATGPT_OAUTH_CLIENT_SECRET`
 - **Authorization URL:** `https://<your-mcp-domain>/authorize`
 - **Token URL:** `https://<your-mcp-domain>/token`
 - **Scope:** `drive` | **Token Exchange:** Default (Basic Auth or Post Body)
- Click **Save & Create MCP**. ChatGPT will verify the manifest and load all 32 Drive tools.

STEP 4 — AUTHORIZE CHATGPT SESSION & LINK GOOGLE DRIVE

- 1. ChatGPT will open the consent dialog. Click **Authorize & Connect** to bind your ChatGPT user session.
- 2. Open a new chat in ChatGPT and invoke your connector:

User: @Google Drive fetch my drive

ChatGPT: Google Drive is not connected for your account. Open this secure one-time link to connect your Google account:
https://<your-mcp-domain>/auth/google/link?code=glink_4f8a912e...
(This link is single-use, expires in 10 minutes, and cryptographically pairs your Drive to this session).

- 3. Click the link → Select your Google Account on the official Google OAuth page → Click **Allow**.
- 4. You will see **"Google Drive Connected! You can now close this tab and return to ChatGPT."**
- 5. Return to ChatGPT and repeat your prompt: **"@Google Drive fetch my drive"**. ChatGPT will now list and manage your Drive files in real time!

3. Search & Discovery Tools

Search your entire Google Drive with keywords, exact filenames, structured filters, or immediate content retrieval:

Tool Name	Access	Description & Capabilities	Example Conversational Prompt
drive_search	Read-Only	Search Drive with simple text queries or structured parameters (name, mime_type, owner_email, dates, parents).	"Search my Drive for all PDFs modified this month"
drive_advanced_search	Read-Only	Multi-clause query constructor supporting complex boolean logic and structured file attribute filtering.	"Find spreadsheets owned by finance with 'Budget' in title"
drive_list_folder	Read-Only	Browse items inside a specific folder with child hierarchy and pagination.	"List all items inside the 'Marketing Q3' folder"
drive_get_metadata	Read-Only	Retrieve full file metadata: ID, MIME type, size, owner, creation/modification dates, sharing state, and permissions.	"Show me full file details and owner for 'PitchDeck.pdf'"
drive_search_and_read	Read-Only	Finds the first matching file for a search phrase and immediately returns its full text content in a single operation.	"Search and read the latest 'Meeting Minutes' document"

4. Reading, Downloading & Exporting Tools

Stream file contents directly to ChatGPT memory with automatic Workspace format conversion:

Tool Name	Access	Description & Capabilities	Example Conversational Prompt
drive_read_file	Read-Only	Read content of text files or automatically export native Google Docs/Sheets/Slides into readable text formats.	"Read the content of 'Executive Summary.txt'"
drive_download_file	Read-Only	Download files or export Docs/Sheets/Slides into PDF, DOCX, CSV, or XLSX formats with binary-safe base64 streaming.	"Download 'Quarterly Review' as a PDF"

5. Native Google Docs API Tools

Read, author, format, and batch-edit native Google Docs without corrupting document structure:

Tool Name	Access	Description & Capabilities	Example Conversational Prompt
drive_doc_create	Read/Write	Create a new native Google Doc with an initial title and optional parent folder.	"Create a new Google Doc named 'Project Roadmap 2027'"
drive_doc_read	Read-Only	Inspect native Google Docs document elements, paragraphs, headings, and full text body via Docs API.	"Read the structure and headings of my 'Design Spec' doc"
drive_doc_update	Read/Write	Execute structured Docs API updates (insertText, replaceAllText, format text, bullets, styles).	"Replace all mentions of 'ACME' with 'Globex' in my doc"
drive_docs_batch_update	Read/Write	Batch update Google Docs with multiple simultaneous editing requests in a single transaction.	"Insert a table and format the title heading in 'Report'"
drive_doc_append	Read/Write	Safely append paragraphs or bullet points to the end of an existing Google Doc.	"Append the meeting conclusion paragraph to 'Client Notes'"

6. Native Google Sheets API Tools

Manipulate spreadsheet cells, read specific A1 ranges, and append tabular records:

Tool Name	Access	Description & Capabilities	Example Conversational Prompt
drive_sheet_create	Read/Write	Create a new Google Spreadsheet with custom sheet titles and grid setup.	"Create a new Google Sheet named 'Q4 Expense Tracker'"
drive_sheet_read_range	Read-Only	Read cell values from an A1 notation range (e.g. Sheet1!A1:D10).	"Read cells A1 to F20 from 'Financial Model'"
drive_sheet_update_range	Read/Write	Write matrices of data directly into a specific A1 range of an existing Sheet.	"Update cells B2:C5 in 'Inventory' with new pricing data"
drive_sheet_append_rows	Read/Write	Append new tabular records to the next available rows in a sheet table.	"Append new customer sign-up rows to 'Leads Sheet'"

7. Native Google Slides Presentation Tools

Create presentation decks, inspect slide structure, and update slide content:

Tool Name	Access	Description & Capabilities	Example Conversational Prompt
drive_slides_create	Read/Write	Create a new Google Slides presentation file with an initial title.	"Create a presentation named 'Investor Pitch Deck'"
drive_slides_read	Read-Only	Inspect slide hierarchy, shapes, speaker notes, and text elements inside a presentation.	"Summarize the slide titles and text in 'Q3 Strategy.pptx'"
drive_slides_update	Read/Write	Execute batch slide modifications (createSlide, insertText, format shapes).	"Add 3 slides with title headings to my pitch presentation"

8. File Management, Organization & Safe Deletion Tools

Create, organize, move, and safely manage your Google Drive files with built-in guardrails:

Tool Name	Access	Description & Capabilities	Example Conversational Prompt
drive_create_file	Read/Write	Upload a new plain-text, markdown, JSON, or CSV file to your Drive.	"Create a new file called 'summary.md' with meeting notes"
drive_create_folder	Read/Write	Create new organizational folders in Drive with optional parent folders.	"Create a folder called 'Archived Projects 2026'"
drive_update_file	Read/Write	Update content of non-native text files (blocked on native Docs to prevent corruption).	"Update 'config.json' with new server parameters"
drive_rename_file	Read/Write	Safely rename any file or folder in Google Drive.	"Rename 'Draft_v1.txt' to 'Final_Document.txt'"
drive_move_file	Read/Write	Relocate files or folders into a new destination folder.	"Move 'Invoice_402.pdf' into the 'Accounting' folder"
drive_copy_file	Read/Write	Duplicate existing files in Drive with custom names and target folders.	"Create a backup copy of 'Contract_Master.docx'"
drive_trash_file	Trash (Safe)	Move files to Google Drive Trash (soft delete). Fully recoverable at any time.	"Move 'Outdated Proposal.pdf' to trash"
drive_restore_file	Restore	Restore a previously trashed file back to active Drive status.	"Restore 'Outdated Proposal.pdf' from trash"
drive_delete_file_permanently	Permanent	Bypasses trash and permanently deletes a single explicit fileId. No retries (maxAttempts: 1).	"Permanently delete file 'temp_cache.log' (irreversible)"

9. Permission Governance & Access Auditing Tools

Inspect and configure file sharing while enforcing strict security boundaries:

Tool Name	Access	Description & Capabilities	Example Conversational Prompt
drive_list_permissions	Read-Only	Audit who has access to a file, including user roles and sharing types.	"Who has access to the 'Confidential Strategy' doc?"
drive_add_permission	Governance	Share file with reader, commenter, or writer roles (ownership transfer blocked).	"Share 'Pitch.pdf' as viewer with teammate@example.com"
drive_update_permission	Governance	Change an existing permission role (e.g. from writer to reader).	"Change Sarah's permission on 'Spec' to commenter only"
drive_remove_permission	Revoke	Revoke file access from a specific user or group.	"Revoke sharing access for contractor@external.com"

Executive Trust & Safety Guarantee:
Google Drive MCP operates under a strict **Zero-Persistence Architecture**. Your personal files, document bodies, spreadsheets, slide presentations, folders, and Google Drive contents are **NEVER saved, written to disk, indexed, cached, or harvested** on our servers.

10. The Five Pillars of User Data Protection

1. Zero File Content Persistence
Data streams directly through volatile runtime memory (RAM) over encrypted TLS/HTTPS and is **immediately purged from memory** once forwarded to your chat session. A hard 10 MB ceiling prevents memory exhaustion.

2. Multi-User Cryptographic Isolation
Every ChatGPT user connects their own Google Account. Internal opaque subjects (`usr_...`) isolate every request. User A can never discover, read, modify, or delete User B's files.

3. AES-256-GCM Storage Encryption
Stored OAuth tokens are protected with AES-256-GCM authenticated encryption, 96-bit random IVs, and restricted POSIX `0600` permissions (never web-accessible).

4. Single-Use Hashed Linking Tokens
Account linking tokens (`glink_...`) are hashed with SHA-256, single-use, and expire in 10 minutes. Tokens are never exposed in browser URLs, query parameters, or client-facing responses.

11. Destructive Safeguards & Guardrail Architecture

- Built-in Safety Guardrails:**
- Trash vs. Permanent Delete:** Standard deletions route safely to Trash (`drive_trash_file`) and are fully reversible. Permanent deletion requires explicit tool selection and confirmation.
 - Non-Retry Rule on Deletion (SEC-05):** Unlike read queries which retry network hiccups, permanent deletion enforces `maxAttempts: 1` so dropped packets never cause repeated deletion attempts.
 - Ownership Transfer Strictly Blocked:** Any request attempting to assign the `owner` role is rejected with an explicit security error (`OWNERSHIP_TRANSFER_BLOCKED`).

12. Instant User Disconnect & Revocation

You maintain absolute ownership and control over your Google credentials at all times:

Method A: Inside ChatGPT
Ask ChatGPT: *"Disconnect my Google Drive account"*. The server immediately calls `POST /auth/google/disconnect`, revokes the token with Google, and permanently deletes local stored credentials.

Method B: Google Account Security
Visit [Google Account Security](#) → [Third-Party Apps](#) and click *Remove Access*. All subsequent requests fail immediately and trigger automatic local credential purge.

13. Compliance & Frequently Asked Questions (FAQ)

Is my data used to train AI models?
No. Data is accessed strictly on demand to fulfill conversational tool requests. The service adheres strictly to the **Google API Services User Data Policy**, including Limited Use requirements.

What if a one-time connection link expires?
Connection links expire after 10 minutes or upon first click. If a link expires, simply ask ChatGPT to perform any Drive search again; a fresh, valid link will be generated automatically.